HK IMO Training 2019-2020
Problem Set 23
For 18 April 2020

Problem 89. Find all ordered pairs of positive integers (m, n) such that

$$m^3 + n^3 = m^2 + 42mn + n^2.$$

Problem 90. Find all injective functions $f\colon \mathbb{R} \to \mathbb{R}$ such that for every real number x and every positive integer n,

$$\left|\sum_{m=1}^{n} m\left(f(x+m+1) - f(f(x+m))\right)\right| < 2020.$$

Problem 91. Let $\mathbb{R}^2 = \{(x, y)\colon x, y \in \mathbb{R}\}$. A *lattice point* is a point with integer coordinates and a *jump vector* is the vector produced after a jump.

A grasshopper is sitting on a lattice point in $\mathbb{R}^2$. Each second it jumps to another lattice point in such a way that the jump vector is constant. A hunter that knows neither the starting point of the grasshopper nor the jump vector (but knows that the jump vector for each second is constant) wants to catch the grasshopper. Each second the hunter can choose one lattice point in $\mathbb{R}^2$ and, if the grasshopper is there, he catches it. Can the hunter always catch the grasshopper in a finite amount of time?

Problem 92. Let ABC be acute-angled triangle with circumcentre O, and let D, E and F be the midpoints of BC, CA and AB, respectively. Let $M \neq D$ be an arbitrary point on BC. Suppose that the straight lines AM and EF intersect at point N and that the straight line ON cuts again the circumcircle of $\triangle ODM$ at point P. Prove that the reflection of point M with respect to the midpoint of DP lies on the nine-point circle of $\triangle ABC$.